Thermal Conductivity Mapping of Interconnects and Active Layers of Logic Chips

Yiyun He
School of Integrated Circuits, Peking University, Beijing, China

Zifeng Huang
School of Integrated Circuits, Peking University, Beijing, China

Ming Li
School of Integrated Circuits, Peking University, Beijing, China

Runsheng Wang
School of Integrated Circuits, Peking University, Beijing, China

Zhe Cheng
School of Integrated Circuits, Peking University, Beijing, China
Email: zhe.cheng@pku.edu.cn

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***Abstract*—With the development of three-dimensional integrated circuits (3D ICs), thermal management has become a critical challenge due to the poor thermal conductivity (TC) of nanoscale interconnects. This study investigates the thermal transport properties of interconnect layers in 16 nm and 5 nm technology nodes using time-domain thermoreflectance (TDTR) and finite element modeling (FEM). The results demonstrate that the effective TC of the interconnect stack is predominantly constrained by thermal boundary conductance (TBC) rather than the intrinsic properties of the constituent materials. The active layer also exhibits a low thermal conductivity even below 1 $W \cdot m^{-1} \cdot K^{-1}$, creating a major bottleneck for heat dissipation in 3D ICs.**

Keywords—TDTR, mapping, thermal conductivity, 3D ICs

I. Introduction

The relentless demand for enhanced integration density and computational performance is propelling the evolution of integrated circuits towards 3D stacking paradigms, spanning wafer-level, die-level, and transistor-level implementations [1,2,3]. Fabrication of 3D ICs generally necessitates substrate thinning [3,4] to minimize interconnect distances, thus reducing signal propagation delay and achieving high packaging density. This structural constraint effectively eliminates lateral heat diffusion paths from the active layers, funneling the generated thermal energy predominantly into the vertical interconnect structure [5,6]. Hence, the thermal transport properties of the interconnects, which consists of metallic and dielectric components, are of great importance in managing the thermal accumulation encountered in stacked device technology [7].

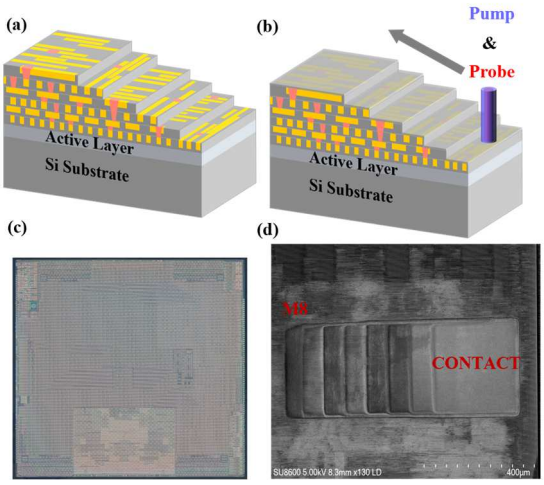


Fig. 1. Schematic diagrams of (a) the step structure fabricated by pFIB and (b) TDTR measurement after the deposition of the Al film. (c) optical micrograph of the 16nm process die and (d) SEM image of the prepared sample.

In this work, the interconnect of a 16 nm and a 5 nm test-chip were processed to expose their cross-sections, enabling a layer-by-layer scan to determine the TC distribution. TDTR mapping, a methodology of paramount importance in the thermal management of micro- and nano-electronic devices [8], was employed for this high-spatial-resolution thermophysical characterization. Combining the TC distribution with the geometrical dimensions of Cu lines within each interconnect layer, as identified by TEM, an FEM representing the 11-layer interconnect stack of a 5 nm chip was developed. The effective TC of the interconnect model is effected by the TC of the Cu lines, dielectric materials, and the TBC of their interfaces.

II. Experiments

A. Sample Preparation

A plasma-based focused ion beam (pFIB) was employed to expose interconnect layers M11 down to M1 in a stepped configuration. Fig. 1(a) illustrates the sample preparation process. After depositing an 80 nm-thick aluminum transducer film on the surface, TDTR experiments were carried out from the interconnect layer closest to the active layer upward, allowing for the construction of a multilayer heat transfer model. The overall experimental procedure is schematically shown in Fig. 1(b). Fig. 1(c) presents an optical microscope image of an unpackaged 16 nm process die, while Fig. 1(d) shows the stepped morphology of the same sample observed by SEM. From the contact layer leftward, the layers are sequentially M1 to M7, with the outermost layer being M8.

Fig. 2 presents the TEM image and EDS spectrum of the 16 nm process chip's interconnect layers. The volumetric heat capacity of the interconnects was estimated by analyzing the elemental composition and concentration.

B. TDTR Mapping on the Interconnect Layers

Fig. 3 schematically illustrates TDTR setup employed in this study. Single-point measurements were initially performed on the interconnect stack. To minimize the impact

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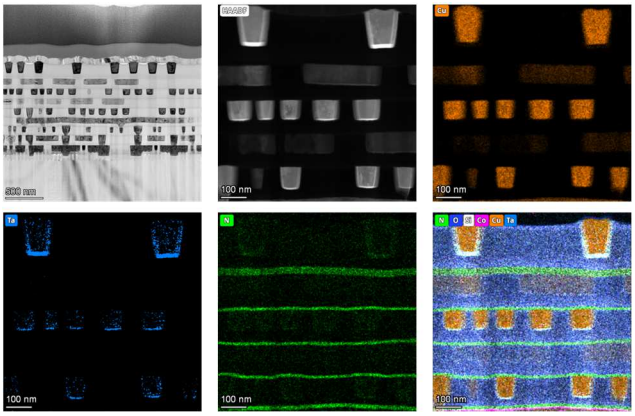


Fig. 2. (a) Cross-sectional TEM image and (b) corresponding EDS elemental maps of the interconnect layers in the 16 nm process chip.

of underlying layers and reduce experimental complexity, two adjacent interconnect layers with similar structures were modeled as a combined unit for measurement. The single-point measurement results for each combined layer pair showed consistency within an 8% error under two laser spot radii (r) configurations (6 μm and 4.8 μm for both pump and probe beams), as evidenced in Fig. 4(a).

Fig. 4(b) plots the sensitivity coefficients of underlying layers TC and TBC of Al/M11M10 as functions of delay time. TC of subjacent layers exhibits negligible sensitivity. The delay time corresponding to near-zero sensitivity of the Al/target-layer TBC was selected for TDTR mapping. At this optimal delay, variations in the TDTR ratio signal directly reflect the cross-plane TC of the measured layer stack.

The spatial in-plane resolution of TDTR mapping, corresponding to the minimum mapping step size, is defined as $\sqrt{2}\,r$. Fig. 5 presents the mapping results of the 16 nm and 5 nm interconnect stacks, obtained using step sizes of 9 μm and 5.5 μm, respectively. As shown in Fig. 5(a), the TDTR ratio distribution of the 16nm interconnects exhibits systematic variation. The ratio increases from the lower layers near the active layer to the upper layers, indicating higher TC in the upper structures. The distribution within individual layers remains relatively uniform.

Fig. 5(b) presents the TC distribution of the 5 nm interconnects, showing a similar trend. Except for local deviations caused by surface defects or contamination, the cross-plane TC of the stack remains below 3 $W \cdot m^{-1} \cdot K^{-1}$. TEM results further show that below the M5M4 level, continued Cu line scaling induces strong interfacial scattering, reducing TC to even below 1 $W \cdot m^{-1} \cdot K^{-1}$. This severely limits vertical heat dissipation from the active layer.

III. Results and Discussion

A comparison of the mapping results obtained with step sizes of 9 μm and 5.5 μm reveals greater intra-layer heterogeneity in the 5.5 μm mapping images. TC distribution correlates with the local interconnect structure. For instance, the mapping results of the M7M6 and M5M4 layers in Fig. 5(b) exhibit localized regions of elevated TC, likely attributable to variations in Cu line density.

Submicron structural features such as via arrangements and Cu line distribution can induce local variations in the cross-plane TC of the interconnect stack. Nevertheless, the overall cross-plane TC remains low. Consequently, layout optimization may offer limited effectiveness in mitigating the

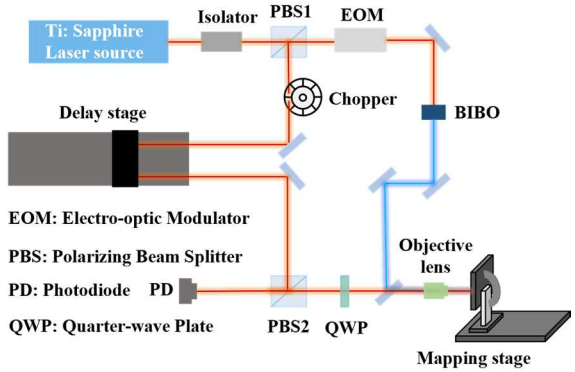


Fig. 3. The schematic diagrams of TDTR.

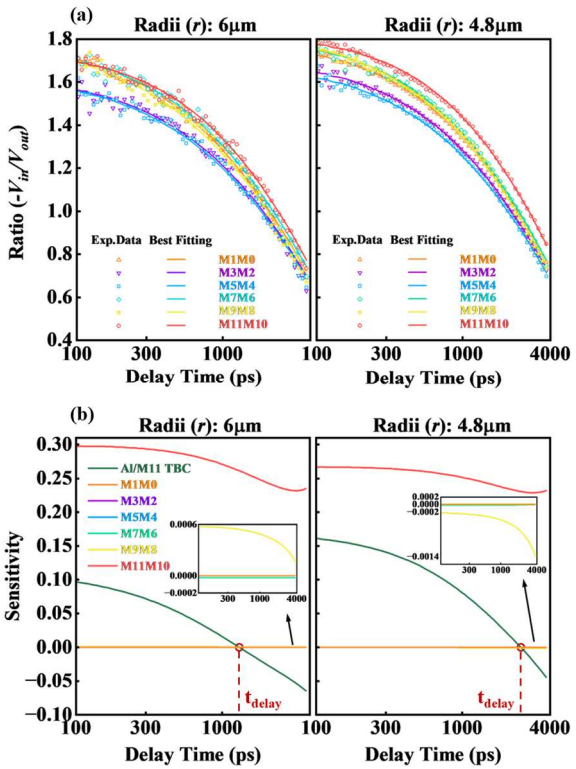


Fig. 4. For the 5 nm interconnect stack, under r of 6 μm (left) and 4.8 μm (right): (a) Single-point TDTR ratio data and the best fitting results, (b) sensitivity coefficients for the TC of each layer and TBC at the Al/M11M10 interface in the measurement on M11M10.

severe thermal dissipation challenges arising from the intrinsically low TC of the interconnect structure.

A pronounced reduction in the TDTR ratio is observed for M0 & AL (active layer) in Fig. 5(a) compared to overlying layers. In contrast, the TC of the M1M0 layer in Fig. 5(b) shows no significant deviation from upper interconnect levels. This apparent discrepancy may originate from two structural factors. First, the active layer contains high-κ dielectrics and exhibits greater material heterogeneity, promoting interfacial scattering of thermal carriers at material boundaries. Second, the Fin-FET architecture introduces complex interfacial geometries through its fin structures, creating additional phonon scattering pathways that further suppress thermal transport in the active layer.

To evaluate the impact of interfacial thermal conductance on the effective TC of the interconnect stack, a FEM was developed. The 11-layer model represents interconnect layers M1 to M11 in the 5 nm process node, with the thickness of

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each layer, as well as the width, height, and pitch of the Cu lines obtained from TEM characterization. Fig. 6(a–c) presents the temperature rise distributions for Cu/dielectric TBC values of 10, 30, and 100 MW/m²K, respectively. Parametric studies were conducted by simultaneously varying the TC of Cu lines, which is modulated by size effects, and the Cu/dielectric TBC.

The results indicate that for TBC values below 80 MW/m²K, the effective TC shows negligible dependence on the copper's intrinsic TC. This suggests that under high interfacial thermal resistance (the reciprocal of TBC), interfacial quality becomes the dominant factor governing the system's effective TC. Based on the experimental findings, enhancing the TBC from 10 to 100 MW/m²K represents the most cost-effective range for improving the interconnect stack's overall thermal performance.

IV. Conclusion

This study investigates the thermal properties of interconnect layers in 16 nm and 5 nm process node chips using both TDTR single-point and mapping measurements. TC mapping results obtained with 9 μm and 5.5 μm step sizes show good consistency, revealing that most interconnect layers exhibit a TC comparable to that of SiO_2. The active layer, containing numerous internal interfaces, demonstrates an even lower TC below 1 $W \cdot m^{-1} \cdot K^{-1}$, indicating severe challenges in dissipating heat vertically from the active region.

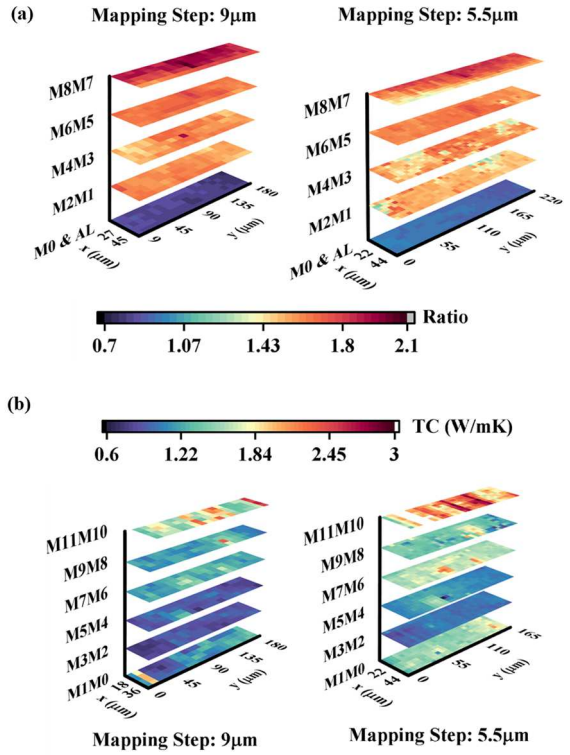


Fig. 5. 3D maps of (a) TDTR ratio signal for the 16 nm interconnects and (b) TC for the 5 nm interconnects, acquired using step sizes of 9 μm (left) and 5.5 μm (right).

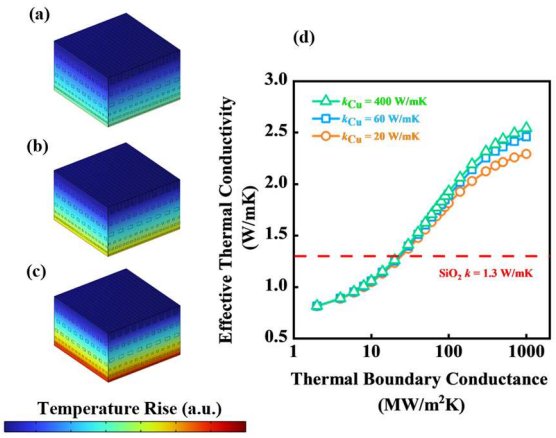


Fig. 6. Finite element analysis results. Temperature rise distributions for Cu/dielectric TBC values of (a) 10, (b) 30, and (c) 100 MW/m²K. (d) Sensitivity of the interconnect stack's effective TC to Cu TC and Cu/dielectric TBC.

FEM analysis identifies TBC as the dominant factor limiting the effective TC of the interconnect stack. Improving interfacial quality and implementing higher-TC materials present promising pathways for mitigating thermal challenges in 3D ICs.

Acknowledgment

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